

(1) State-guided Control

Research Checkpoint

Before next tool query

(e.g. search/open/ find)

State Signals

Confidence C^t

Evidence Attention A^t

Convergence D^t

Trained
Reroute
Predictor



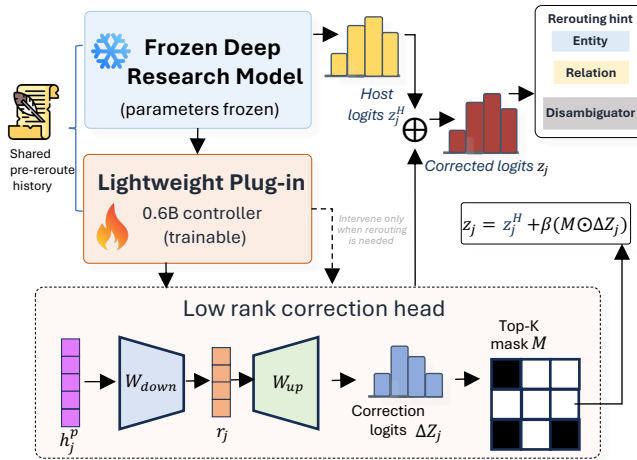
CONTINUE

or STOP

or REROUTE

Trigger logit
intervention

(2) Logit Intervention



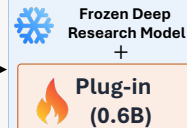
(3) Optimization and inference

Optimization:



BrowseCore

Question + local corpus



Output

Reward
function

Inference:

Frozen Deep
Research Model

State
Control

Trigger
Reroute

0.6B controller

Rerouted
Hint

stop or
continue

Turn n